

Write a Java program to read and simulate the execution of a program written in a highly simplified programming language. Each line in the input contains one statement. There are no declaration statements; all values are of type *double*. Statements in the language are in the following forms:

Simple Statements	Meaning
<i>variable</i> = <i>expression</i>	The <i>expression</i> consists of variables, constants, and the simple arithmetic operators (+, -, *, /) with blanks as delimiters. It is evaluated left to right and the result is placed in the <i>variable</i> . Note that there is no hierarchy among the operators and parentheses are not allowed.
PRINT <i>variable</i>	The value of the <i>variable</i> should be displayed with two digits to the right of the decimal point.
GOTO <i>n</i>	Execution moves to the <i>n</i> th line of the program. The first line in the program is line 1.

Conditional Statement	Meaning
IF <i>variable</i> IS <i>value</i> THEN <i>simpleStatement</i>	If the <i>variable</i> is equal to <i>value</i> (a variable or a constant), the <i>simpleStatement</i> is performed.

End of Program Marker	Meaning
END	This is the last statement in a program. There are no statements following this. When execution reaches this statement, the program is terminated.

PRINT, GOTO, IF, and END are reserved words; i.e., they cannot be used as variable names.

The GUI will include a “Next Program” button which allows the user to select a file containing a program to be executed. The program in the selected file is then executed and the output (result of the PRINT statements) should be displayed in a scrollable text area.

NOTE

Use of an array is NOT acceptable in this assignment.

As in Lab 1, you will be given code for the GUI.

Required elements:

- Error checking and reporting
- Good choice of variable names
- Well structured & documented